

Saketh Maddineni

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SUMMARY

Python Developer with 4+ years of experience delivering scalable backend and full-stack solutions across healthcare and enterprise domains. Skilled in Python, Django, Flask, REST APIs, and modern front-end frameworks like React and Angular. Proven success in building HIPAA-compliant data systems, automating ETL pipelines, and deploying microservices on cloud platforms including AWS and Azure. Adept at cross-functional collaboration in Agile teams, with a strong focus on clean code, automation, and performance optimization.

SKILLS

Programming Languages: Python (including Python 3.x, Pandas, NumPy, SciPy), JavaScript, C, Java, Erlang, Prolog

Web & API Development: Django, Flask, FastAPI, RESTful APIs, GraphQL APIs, React.js, AngularJS, HTML5, CSS3, Bootstrap

Database Systems: MySQL, PostgreSQL, Microsoft SQL Server, SQLiteNoSQL, MongoDB, Oracle, ChromaDB

Cloud & Infrastructure: AWS (EC2, S3, RDS), Azure, Google Cloud Platform (GCP), Terraform

DevOps & CI/CD Tools: Docker, Kubernetes, Jenkins, Ansible, Git

Tools & Utilities: JIRA, Confluence, Bitbucket, HPQC, Tableau, FuelTrak

IDEs & Development Tools: VS Code, PyCharm, IntelliJ, Eclipse, Atom, Spyder

Methodologies: Agile, Scrum, Test-Driven Development (TDD), Version Control (Git)

EXPERIENCE

Software Development Engineer

Humana Inc.

July 2024 - Present, USA

- Developed and deployed a secure, **HIPAA-compliant patient data ingestion system** using **Python, Django REST Framework**, and **FHIR standards**, enabling real-time access to **EHRs** and reducing data retrieval latency by **30%**.
- Led integration of a microservices-based architecture on **AWS**, incorporating **S3, Lambda, EC2**, and **API Gateway**, resulting in 99.9% uptime and scalable access across 5 departments.
- Automated healthcare claims validation workflows using **Pandas, Sklearn**, and business rules logic, resulting in a **95%** accuracy rate and saving an estimated **120+ manual hours/month** for operations teams.
- Designed and developed a full-stack predictive analytics dashboard using **Python (Flask), React**, and **Tableau**, collaborating with data scientists and QA to visualize treatment outcomes and reduce patient readmission rates by **15%**.
- Led a successful data migration initiative of over **5 million medical records** from legacy systems to **PostgreSQL**, ensuring zero data loss and **100%** compliance with regulatory requirements through rigorous **QA** and **Pytest** coverage.

Backend Python Developer

Tata Consultancy Services

January 2021 - July 2023, India

- Architected and deployed robust backend services using **Python, Django**, and **REST APIs** to integrate patient health records with custom healthcare platforms, ensuring seamless data synchronization across client-facing applications.
- Engineered a scalable microservices architecture, employing **Docker** and **Kubernetes** to automate deployment processes, which improved system efficiency by **20%** and reduced downtime by **30%**.
- Collaborated with front-end teams to build responsive user interfaces using **React, Angular, HTML5**, and **CSS3**, reducing UI-related bug tickets by 30%.
- Actively participated in Agile sprint planning and cross-functional meetings, using **JIRA, Confluence**, and **Bitbucket** to manage code reviews, documentation, and collaborative development workflows.

Software Engineer

Tata Consultancy Services

May 2020 - December 2020, India

- Designed and deployed **RESTful APIs** using **Python** and **Django** to support high-throughput fuel management systems, handling over **2TB** of data and reducing request latency by **20%** through optimized database access and caching strategies.
- Streamlined data migration for defect tracking environments, implementing custom scripts and automation with **Python** and **SQL** to ensure compatibility and support for over **10,000+** test cases during system upgrades.
- Developed backend services in **Python** to integrate real-time VIN/PIN tracking into the **FuelTrak** system, increasing operational efficiency by **30%**.

EDUCATION

Master of Science in Computer Science

University of Massachusetts, Boston • Boston, MA • 2025

Bachelors of Technology in Computer Science

National Institute of Technology • India • 2023

PUBLICATIONS

CDADITagger

Document • IEEE • 2022

- Approach Towards Content-Based Annotations Driven Image Tagging Integrating Hybrid Semantics

HyResPR

Conference paper • Springer • 2023

- Hybridized Framework for Recommendation of Research Paper Using Semantically Driven ML Models

A Hybrid Deep Transformer Model for Epileptic Seizure Prediction

Document • IEEE • 2022

Influence Maximization on Socially Scalable Networks using GANs

Document • Bachelor's Thesis • 2023

PROJECT

Movie Ticket Booking Application

University of Massachusetts Boston • December 2024 - March 2025

- Implemented RabbitMQ handling 5K messages/min for workflows, designed GraphQL schemas reduced 40% overfetching.
- Scaled it to handle 1K+ concurrent users, used Redis for distributed locking preventing overbooking & ensure data consistency.

Fine-tuning Large Language Models (LLMs)

University of Massachusetts Boston • January 2023 - May 2023

- Deployed LLaMA 2 on Lambda Labs for scalable, reproducible inference with consistent output and sub-5ms latency.
- Employed quantization, pruning to optimize model size, reducing memory footprint by 70% while retaining performance.
- Accelerated inference latency by 50% through distillation and neural architecture search.

Distributed Task Scheduler Using gRPC

National Institute of Technology Andhra Pradesh • November 2022 - February 2023

- Engineered an ultra-low latency distributed task scheduler using gRPC, achieving sub-5ms latency for 10,000 tasks, dynamic task reassignment, and real-time worker failure detection through shared memory and POSIX locks.
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